

UFUG 2106 : Discrete Mathematics

Chapter 07: Properties of Functions

Outline

1. Functions Defined on General Sets
2. One-to-One, Onto, and Inverse Functions
3. Composition of functions
4. Cardinality with Applications to Computability

Intended Learning Outcomes

By the end of this chapter, students should be able to:

1. Define and work with functions on general sets using diagrams, tables, and formal notation.
2. Determine whether functions are one-to-one, onto, or bijective, and construct or verify inverses when applicable.
3. Perform and analyze compositions of functions, including with identity and inverse functions.
4. Compare the cardinalities of finite, countably infinite, and uncountable sets.
5. Explain implications of cardinality in the context of computability, including why some functions are not computable.

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Functions: Definitions Revisited

Definition

A **function** f from a set X to a set Y , denoted $f: X \rightarrow Y$, is a relation from X , the **domain** of f , to Y , the **co-domain** of f , that satisfies two properties: (1) every element in X is related to some element in Y , and (2) no element in X is related to more than one element in Y .

Thus, given any element x in X , there is a unique element in Y that is related to x by f . If we call this element y , then we say that “ f sends x to y ” or “ f maps x to y ” and write $x \xrightarrow{f} y$ or $f: x \rightarrow y$. The unique element to which f sends x is denoted

$f(x)$ and is called f of x , or
the output of f for the input x , or
the value of f at x , or
the image of x under f .

The set of all values of f taken together is called the *range of f* or the *image of X under f* . Symbolically:

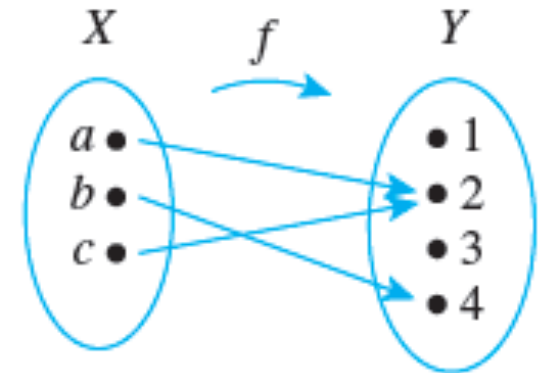
$$\text{range of } f = \text{image of } X \text{ under } f = \{y \in Y \mid y = f(x), \text{ for some } x \text{ in } X\}.$$

Given an element y in Y , there may exist elements in X with y as their image. When x is an element such that $f(x) = y$, then x is called a **preimage of y** or an **inverse image of y** . The set of all inverse images of y is called *the inverse image of y* . Symbolically:

$$\text{the inverse image of } y = \{x \in X \mid f(x) = y\}.$$

Functions: Definitions Revisited

Let $X = \{a, b, c\}$ and $Y = \{1, 2, 3, 4\}$. Define a function f from X to Y by the arrow diagram.



- Write the domain and co-domain of f .
- Find $f(a)$, $f(b)$, and $f(c)$.
- What is the range of f ?
- Is c an inverse image of 2 ? Is b an inverse image of 3 ?
- Find the inverse images of 2 , 4 , and 1 .
- Represent f as a set of ordered pairs.

Functions: Definitions Revisited

- a. domain of $f = \{a, b, c\}$, co-domain of $f = \{1, 2, 3, 4\}$
- b. $f(a) = 2$, $f(b) = 4$, $f(c) = 2$
- c. range of $f = \{2, 4\}$
- d. yes, no
- e. inverse image of $2 = \{a, c\}$
inverse image of $4 = \{b\}$
inverse image of $1 = \emptyset$ (since no arrows point to 1)
- f. $\{(a, 2), (b, 4), (c, 2)\}$

Example: Encoding and Decoding Functions

Digital messages consist of finite sequences of 0's and 1's.

For example, we have the message

00101111

can be encoded as

0000001110001111111111.

Boolean Functions

Definition

An (**n -place**) **Boolean function** f is a function whose domain is the set of all ordered n -tuples of 0's and 1's and whose co-domain is the set $\{0, 1\}$. More formally, the domain of a Boolean function can be described as the Cartesian product of n copies of the set $\{0, 1\}$, which is denoted $\{0, 1\}^n$. Thus $f: \{0, 1\}^n \rightarrow \{0, 1\}$.

Consider the three-place Boolean function defined from the set of all 3-tuples of 0's and 1's to $\{0, 1\}$ as follows: For each triple (x_1, x_2, x_3) of 0's and 1's,

$$f(x_1, x_2, x_3) = (x_1 + x_2 + x_3) \bmod 2.$$

Describe f using an input/output table.

Example: The Hamming Distance Function

The Hamming distance function gives a measure of the “**difference**” between two strings of 0’s and 1’s that have the **same length**. Let S_n be the set of all strings of 0’s and 1’s of length n .

Define a function $H: S_n \times S_n \rightarrow \mathbf{Z}^{nonneg}$ as follows: For each pair of strings $(s, t) \in S_n \times S_n$,

$H(s, t)$ = the number of positions in which s and t have different values.

Letting $n = 5$,

$$H(11111, 00000) = 5$$

because 11111 and 00000 differ in all five positions.

Functions Acting on Sets

Definition

If $f: X \rightarrow Y$ is a function and $A \subseteq X$ and $C \subseteq Y$, then

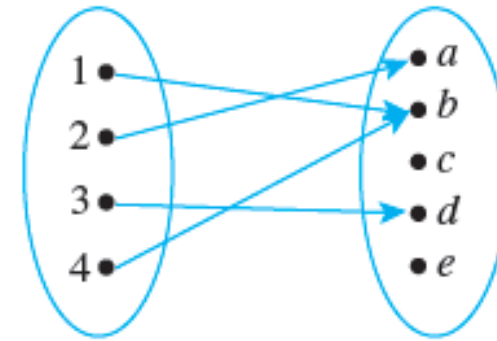
$$f(A) = \{y \in Y \mid y = f(x) \text{ for some } x \text{ in } A\}$$

and

$$f^{-1}(C) = \{x \in X \mid f(x) \in C\}.$$

$f(A)$ is called the **image of A** , and $f^{-1}(C)$ is called the **inverse image of C** .

Let $X = \{1, 2, 3, 4\}$ and $Y = \{a, b, c, d, e\}$, and define $F: X \rightarrow Y$ by the following arrow diagram:



Let $A = \{1, 4\}$, $C = \{a, b\}$, and $D = \{c, e\}$.

Find $F(A)$, $F(X)$, $F^{-1}(C)$, $F^{-1}(D)$.

Functions Acting on Sets

$$F(A) = \{b\}$$

$$F(X) = \{a, b, d\}$$

$$F^{-1}(C) = \{1, 2, 4\}$$

$$F^{-1}(D) = \emptyset$$

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One-to-One Functions

Definition

Let F be a function from a set X to a set Y . F is **one-to-one** (or **injective**) if, and only if, for all elements x_1 and x_2 in X ,

if $F(x_1) = F(x_2)$, then $x_1 = x_2$,

or, equivalently,

if $x_1 \neq x_2$, then $F(x_1) \neq F(x_2)$.

Symbolically:

$F: X \rightarrow Y$ is one-to-one $\Leftrightarrow \forall x_1, x_2 \in X$, if $F(x_1) = F(x_2)$ then $x_1 = x_2$.

A function $F: X \rightarrow Y$ is *not* one-to-one $\Leftrightarrow \exists$ elements x_1 and x_2 in X with
 $F(x_1) = F(x_2)$ and $x_1 \neq x_2$.

Proving or Disproving That Functions Are One-to-One

Define $f: \mathbf{R} \rightarrow \mathbf{R}$ and $g: \mathbf{Z} \rightarrow \mathbf{Z}$ by the rules

$$f(x) = 4x - 1 \text{ for all } x \in \mathbf{R}$$

and

$$g(n) = n^2 \text{ for all } n \in \mathbf{Z}.$$

- a. Is f one-to-one? Prove or give a counterexample.
- b. Is g one-to-one? Prove or give a counterexample.

Application: Hash Functions

Definition: Hash Function

A **hash function** is a function defined from a larger, possibly infinite, set of data to a smaller fixed-size set of integers.

Suppose there are only 6 students. Define a function H , from the set of student ID numbers to the set $\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$ as follows:

$$H(n) = n \bmod 11 \text{ for each ID number } n.$$

As noted, H may not be one-to-one: two different ID numbers could have the same H -value (**collision**).

Application: Hash Functions

One of the simplest, called a **linear probe**, works as follows:

- If $H(n)$ is already occupied when a new student ID number is input, start from $H(n)$ in the table and search downward to put a link for the student's record in the first empty position that occurs;
- if no empty positions remain going down, go up to the beginning of the table and search from there.

Onto Functions

Definition

Let F be a function from a set X to a set Y . F is **onto** (or **surjective**) if, and only if, given any element y in Y , it is possible to find an element x in X with the property that $y = F(x)$.

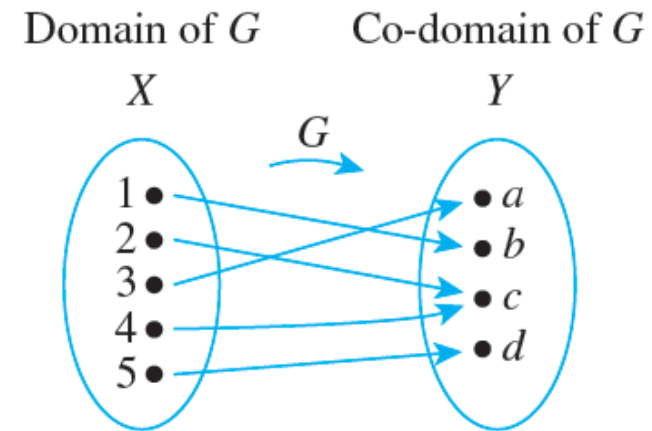
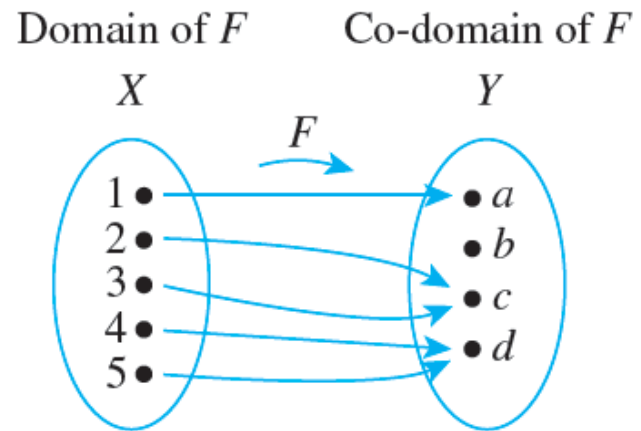
Symbolically:

$$F: X \rightarrow Y \text{ is onto} \iff \forall y \in Y, \exists x \in X \text{ such that } F(x) = y.$$

$$F: X \rightarrow Y \text{ is not onto} \iff \exists y \text{ in } Y \text{ such that } \forall x \in X, F(x) \neq y.$$

Onto Functions

- Do either of the arrow diagrams define onto functions?



Proving or Disproving That Functions Are Onto

Define $f: \mathbf{R} \rightarrow \mathbf{R}$ and $h: \mathbf{Z} \rightarrow \mathbf{Z}$ by the rules

$$f(x) = 4x - 1 \text{ for each } x \in \mathbf{R}$$

and

$$h(n) = 4n - 1 \text{ for each } n \in \mathbf{Z}.$$

- a. Is f onto? Prove or give a counterexample.
- b. Is h onto? Prove or give a counterexample.

One-to-One Correspondences

Definition

A **one-to-one correspondence** (or **bijection**) from a set X to a set Y is a function $F: X \rightarrow Y$ that is both one-to-one and onto.

Inverse Functions

Theorem 7.2.2

Suppose $F: X \rightarrow Y$ is a one-to-one correspondence; in other words, suppose F is one-to-one and onto. Then there is a function $F^{-1}: Y \rightarrow X$ that is defined as follows: Given any element y in Y ,

$$F^{-1}(y) = \text{that unique element } x \text{ in } X \text{ such that } F(x) \text{ equals } y.$$

Or, equivalently,

$$F^{-1}(y) = x \iff y = F(x).$$

Definition

The function F^{-1} is called the **inverse function** for F .

Inverse Functions

The function $f: \mathbf{R} \rightarrow \mathbf{R}$ defined by the formula

$$f(x) = 4x - 1 \text{ for each real number } x$$

was shown to be one-to-one in Example 7.2.2 and onto in Example 7.2.5. Find its inverse function.

Inverse Functions

For any *[particular but arbitrarily chosen]* y in $\mathbf{R}$, by definition of f^{-1} ,

$f^{-1}(y)$ = that unique real number x such that $f(x) = y$.

But

$$f(x) = y$$

$$\Leftrightarrow 4x - 1 = y \quad \text{by definition of } f$$

$$\Leftrightarrow x = \frac{y + 1}{4} \quad \text{by algebra.}$$

$$\text{Hence } f^{-1}(y) = \frac{y + 1}{4}.$$

Inverse Functions

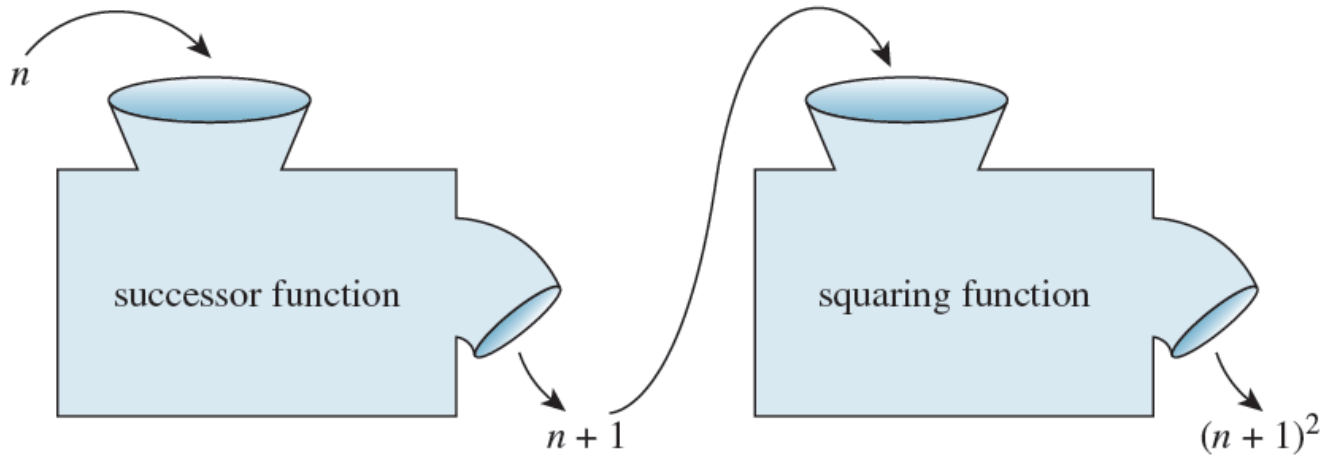
Theorem 7.2.3

If X and Y are sets and $F: X \rightarrow Y$ is one-to-one and onto, then $F^{-1}: Y \rightarrow X$ is also one-to-one and onto.

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Composition of Functions



Definition

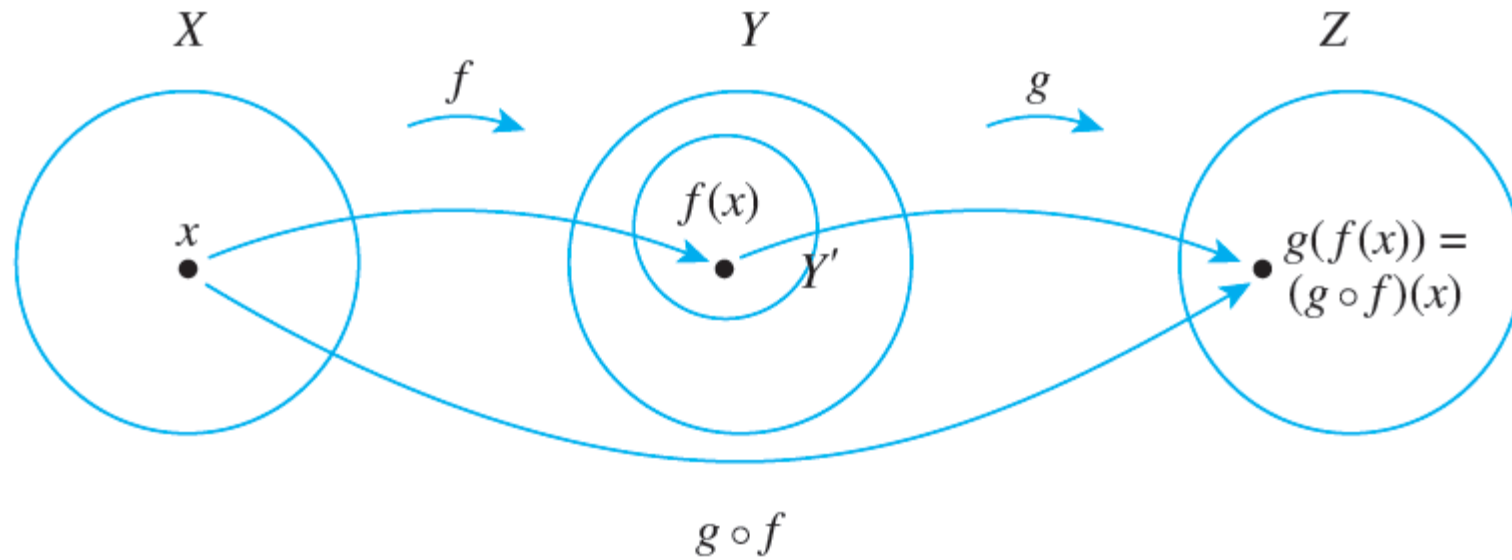
Let $f: X \rightarrow Y$ and $g: Y' \rightarrow Z$ be functions with the property that the range of f is a subset of the domain of g . Define a new function $g \circ f: X \rightarrow Z$ as follows:

$$(g \circ f)(x) = g(f(x)) \quad \text{for each } x \in X,$$

where $g \circ f$ is read “ g circle f ” and $g(f(x))$ is read “ g of f of x .” The function $g \circ f$ is called the **composition of f and g** .

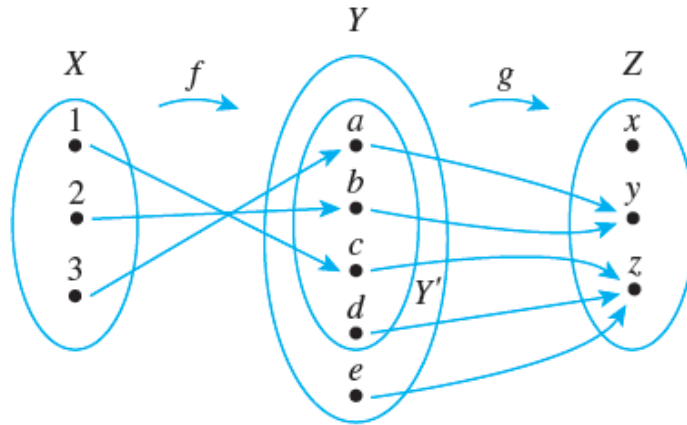
Composition of Functions

This definition is shown schematically below.



Composition of Functions

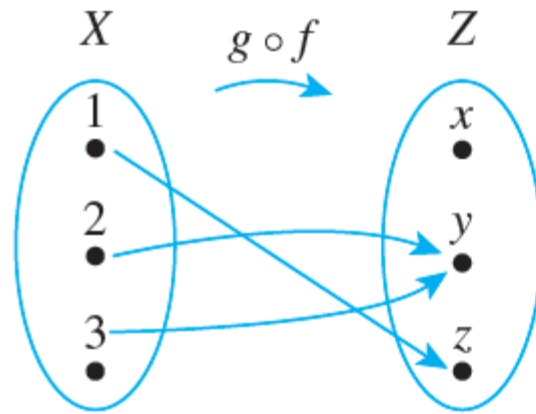
Let $X = \{1, 2, 3\}$, $Y' = \{a, b, c, d\}$, $Y = \{a, b, c, d, e\}$, and $Z = \{x, y, z\}$. Define functions $f: X \rightarrow Y'$ and $g: Y \rightarrow Z$ by the arrow diagrams below.



Draw the arrow diagram for $g \circ f$. What is the range of $g \circ f$?

Composition of Functions

To find the arrow diagram for $g \circ f$, just trace the arrows all the way across from X to Z through Y . The result is shown below.



$$(g \circ f)(1) = g(f(1)) = g(c) = z$$

$$(g \circ f)(2) = g(f(2)) = g(b) = y$$

$$(g \circ f)(3) = g(f(3)) = g(a) = y$$

The range of $g \circ f$ is $\{y, z\}$.

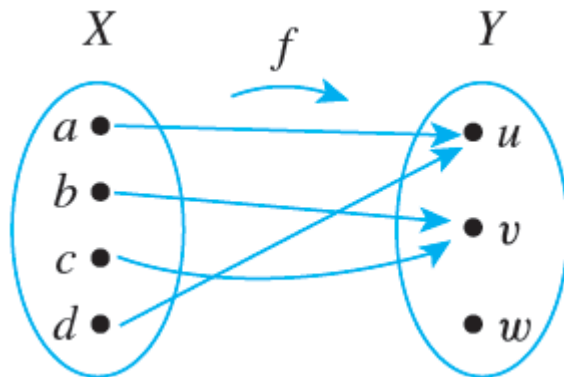
Composition with the Identity Function

Given a set X , define a function I_X from X to X by

$$I_X(x) = x \quad \text{for each } x \text{ in } X.$$

The function I_X is called the **identity function on X** .

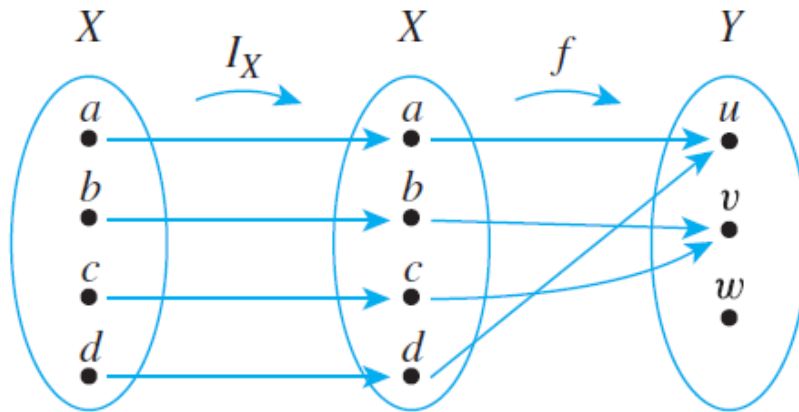
Let $X = \{a, b, c, d\}$ and $Y = \{u, v, w\}$, and suppose $f: X \rightarrow Y$ is given by the arrow diagram shown below.



Find $f \circ I_X$ and $I_Y \circ f$.

Composition with the Identity Function

The values of $f \circ I_X$ are obtained by tracing through the arrow diagram shown below.



$$\begin{aligned}(f \circ I_X)(a) &= f(I_X(a)) = f(a) = u \\(f \circ I_X)(b) &= f(I_X(b)) = f(b) = v \\(f \circ I_X)(c) &= f(I_X(c)) = f(c) = v \\(f \circ I_X)(d) &= f(I_X(d)) = f(d) = u\end{aligned}$$

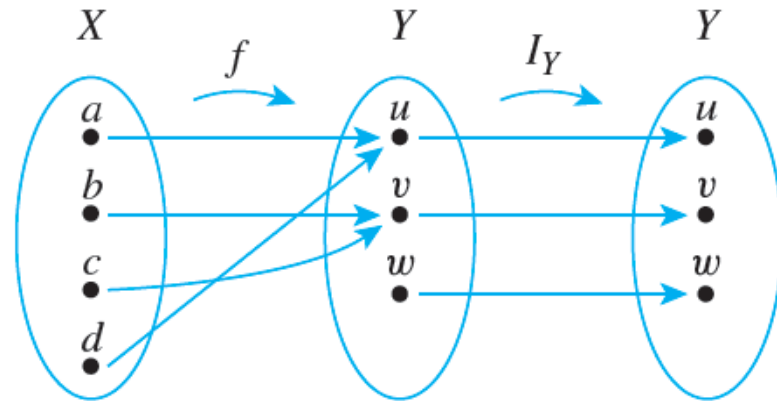
Thus, for every element x in X ,

$$(f \circ I_X)(x) = f(x).$$

By definition of equality of functions, this means that $f \circ I_X = f$.

Composition with the Identity Function

Similarly, the equality $I_Y \circ f = f$ can be verified by tracing through the arrow diagram below for each x in X and noting that in each case, $(I_Y \circ f)(x) = f(x)$.



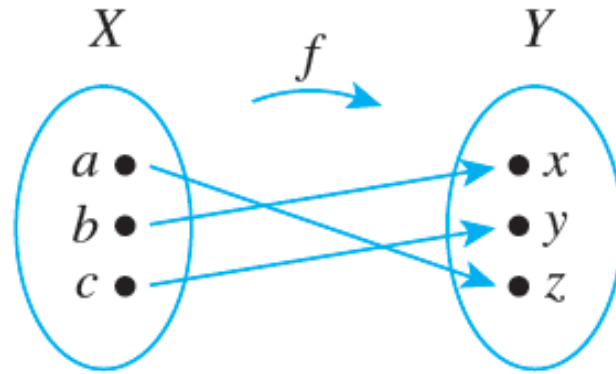
Theorem 7.3.1 Composition with an Identity Function

If f is a function from a set X to a set Y , and I_X is the identity function on X , and I_Y is the identity function on Y , then

$$(a) f \circ I_X = f \quad \text{and} \quad (b) I_Y \circ f = f.$$

Composing a Function with Its Inverse

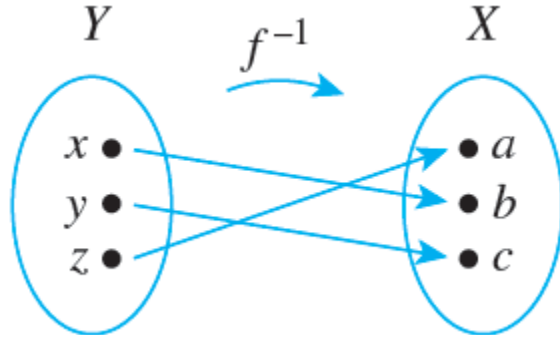
Let $X = \{a, b, c\}$ and $Y = \{x, y, z\}$. Define $f: X \rightarrow Y$ by the following arrow diagram.



You can see from the diagram that f is one-to-one and onto.

Composing a Function with Its Inverse

Thus f^{-1} exists and is found by tracing the arrows backwards, as shown below.



Now $f^{-1} \circ f$ is found by following the arrows from X to Y by f and back to X by f^{-1} .

If you do this, you will see that

$$(f^{-1} \circ f)(a) = f^{-1}(f(a)) = f^{-1}(z) = a$$

Composing a Function with Its Inverse

$$(f^{-1} \circ f)(b) = f^{-1}(f(b)) = f^{-1}(x) = b$$

and

$$(f^{-1} \circ f)(c) = f^{-1}(f(c)) = f^{-1}(y) = c.$$

Thus the composition of f and f^{-1} sends each element to itself. So by definition of the identity function,

$$f^{-1} \circ f = I_X.$$

In a similar way, you can see that

$$f \circ f^{-1} = I_Y.$$

Theorem 7.3.2 Composition of a Function with Its Inverse

If $f: X \rightarrow Y$ is a one-to-one and onto function with inverse function $f^{-1}: Y \rightarrow X$, then

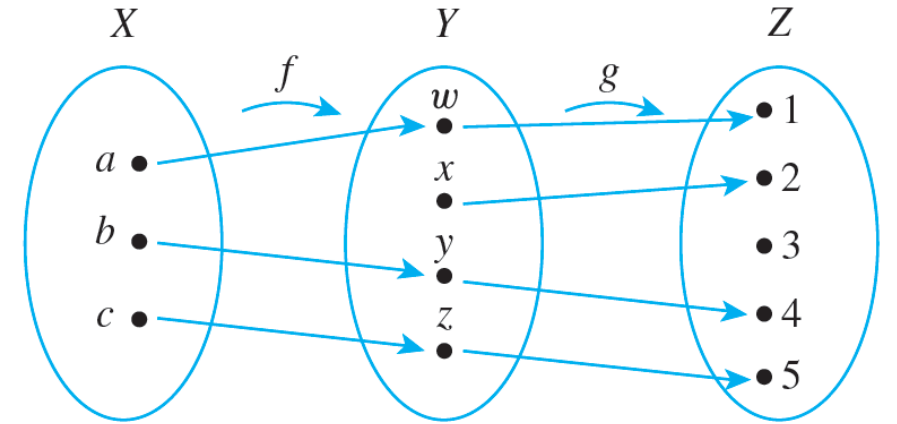
$$(a) f^{-1} \circ f = I_X \quad \text{and} \quad (b) f \circ f^{-1} = I_Y.$$

Composition of One-to-One Functions

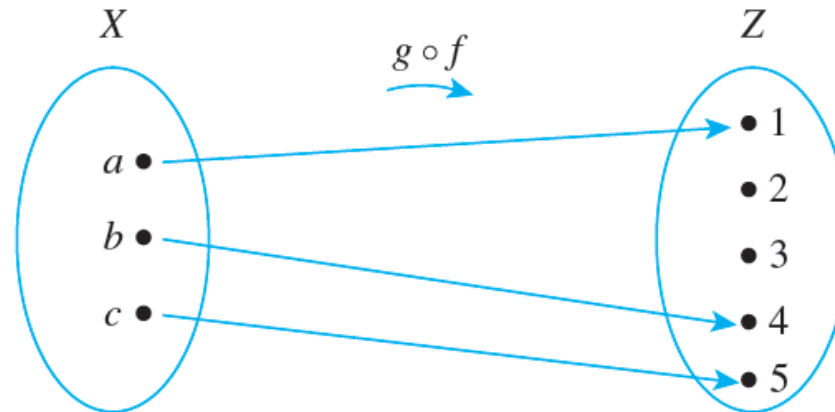
Let

$X = \{a, b, c\}$, $Y = \{w, x, y, z\}$, and $Z = \{1, 2, 3, 4, 5\}$, and define one-to-one functions

$f: X \rightarrow Y$ and $g: Y \rightarrow Z$ as shown in the arrow diagrams.



Then $g \circ f$ is the function with the arrow diagram.



Composition of One-to-One Functions

From the diagram it is clear that for these particular functions, the composition is one-to-one.

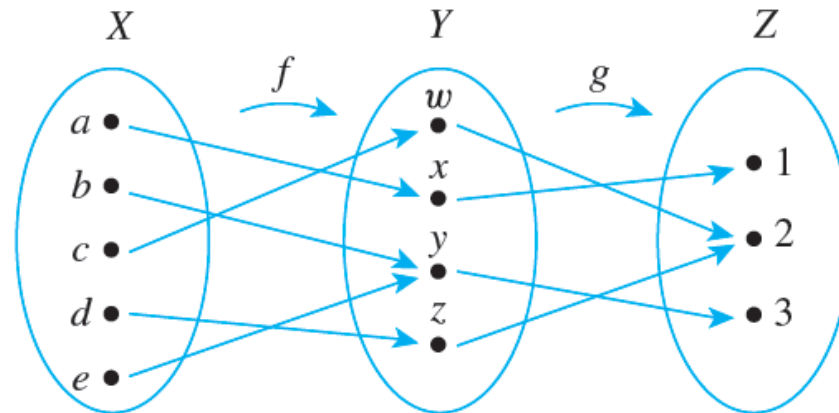
Theorem 7.3.3

If $f: X \rightarrow Y$ and $g: Y \rightarrow Z$ are both one-to-one functions, then $g \circ f$ is one-to-one.

Composition of Onto Functions

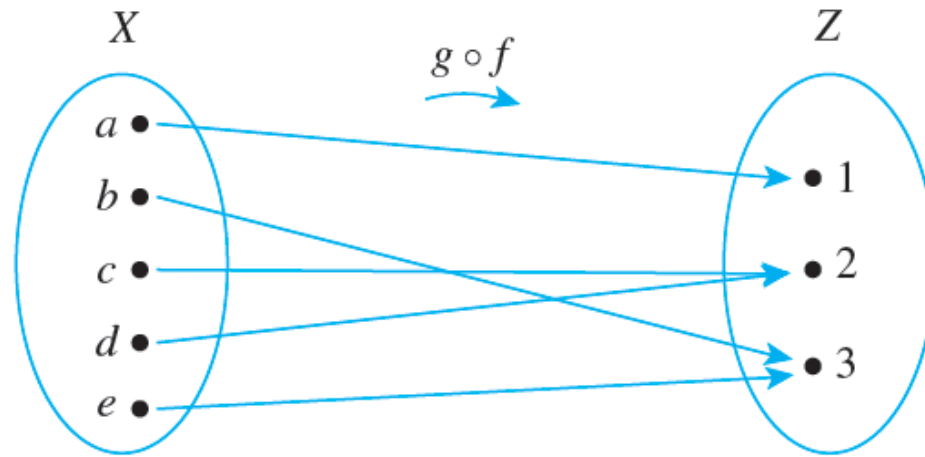
Now consider what happens when two onto functions are composed. For example, let $X = \{a, b, c, d, e\}$, $Y = \{w, x, y, z\}$, and $Z = \{1, 2, 3\}$.

Define onto functions f and g by the following arrow diagrams.



Composition of Onto Functions

Then $g \circ f$ is the function with the arrow diagram shown below.



It is clear from the diagram that $g \circ f$ is onto.

Theorem 7.3.4

If $f: X \rightarrow Y$ and $g: Y \rightarrow Z$ are both onto functions, then $g \circ f$ is onto.

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Cardinality with Applications to Computability

The term ***cardinal number*** refers to the **size of a set** (“This set has *eight* elements”), whereas the term ***ordinal number*** refers to the **order** of an element in a sequence (“This is the *eighth* element in the row”).

Cardinality with Applications to Computability

Definition

Let A and B be any sets. A **has the same cardinality as B** if, and only if, there is a one-to-one correspondence from A to B . In other words, A has the same cardinality as B if, and only if, there is a function f from A to B that is one-to-one and onto.

The following theorem gives some basic properties of cardinality,

Theorem 7.4.1 Properties of Cardinality

For all sets A , B , and C :

- Reflexive property of cardinality:** A has the same cardinality as A .
- Symmetric property of cardinality:** If A has the same cardinality as B , then B has the same cardinality as A .
- Transitive property of cardinality:** If A has the same cardinality as B and B has the same cardinality as C , then A has the same cardinality as C .

Cardinality with Applications to Computability

Definition

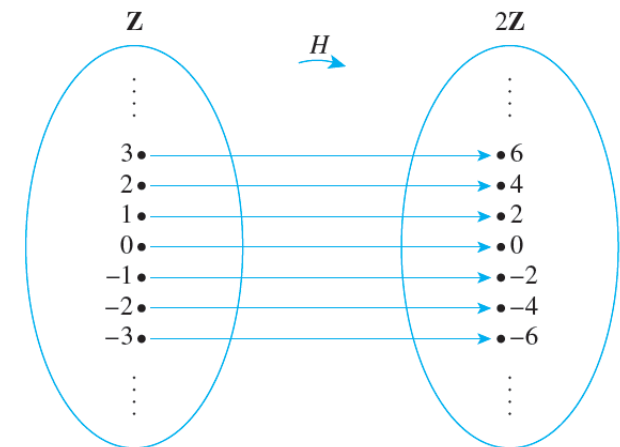
A and B **have the same cardinality** if, and only if, A has the same cardinality as B or B has the same cardinality as A .

Let $2\mathbf{Z}$ be the set of all even integers.

Prove that $2\mathbf{Z}$ and $\mathbf{Z}$ have the same cardinality.

Consider the function H from $\mathbf{Z}$ to $2\mathbf{Z}$ defined as follows:

$H(n) = 2n$ for each $n \in \mathbf{Z}$.



Cardinality with Applications to Computability

To show that H is **one-to-one**, suppose $H(n_1) = H(n_2)$ for some integers n_1 and n_2 .

Then $2n_1 = 2n_2$ by definition of H , and dividing both sides by 2 gives $n_1 = n_2$. Hence h is one-to-one.

To show that H is **onto**, suppose m is any element of $2\mathbf{Z}$. Then m is an even integer, and so $m = 2k$ for some integer k . It follows that $H(k) = 2k = m$.

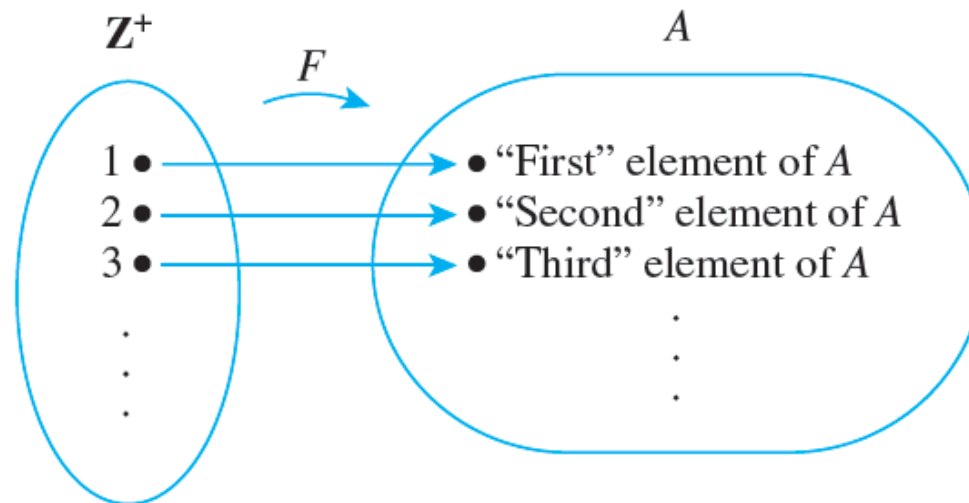
Thus there exists k in $\mathbf{Z}$ with $H(k) = m$, and hence H is onto.

Therefore, by definition of cardinality, $\mathbf{Z}$ and $2\mathbf{Z}$ have the same cardinality.

Countable Sets

Definition

A set is **finite** if, and only if, it is the empty set or can be put into one-to-one correspondence with a set of the form $\{1, 2, \dots, n\}$ for some positive integer n . A set is **countably infinite** if, and only if, it has the same cardinality as the set of positive integers $\mathbf{Z}^+$. A set is **countable** if, and only if, it is finite or countably infinite. A set that is not countable is called **uncountable**.

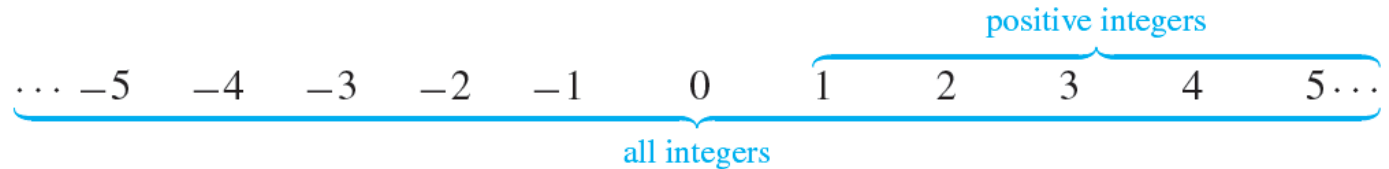


"Counting" a Countably Infinite Set

Countable Sets

Show that $\mathbf{Z}$, the set of all integers, is countable.

The set $\mathbf{Z}$ is certainly not finite, so if it is countable, it must be because it is countably infinite. To show that $\mathbf{Z}$ is countably infinite, find a function from the positive integers $\mathbf{Z}^+$

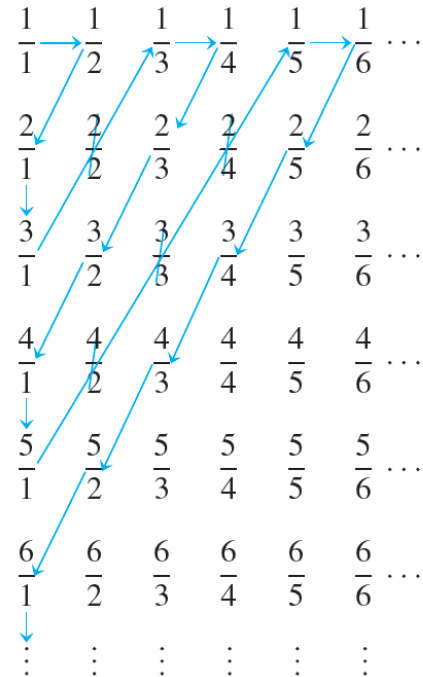


$$F(n) = \begin{cases} \frac{n}{2} & \text{if } n \text{ is an even positive integer} \\ -\frac{n-1}{2} & \text{if } n \text{ is an odd positive integer.} \end{cases}$$

The Set of All Positive Rational Numbers Is Countable

Show that $\mathbb{Q}^+$, the set of all positive rational numbers, is countable.

Display the elements of the set $\mathbb{Q}^+$ in a grid as shown below.



The Set of All Positive Rational Numbers Is Countable

Define a function F from $\mathbf{Z}^+$ to $\mathbf{Q}^+$ by starting to count at $\frac{1}{1}$ and following the arrows as indicated, skipping over any number that has already been counted.

To be specific:

Set $F(1) = 1$, $F(2) = \frac{1}{2}$, $F(3) = \frac{2}{1}$, and $F(4) = \frac{3}{1}$.

Then skip $\frac{2}{2}$ since $\frac{2}{2} = \frac{1}{1}$, which was counted already.

After that, set $F(5) = \frac{1}{3}$, $F(6) = \frac{1}{4}$, $F(7) = \frac{2}{3}$, $F(8) = \frac{3}{2}$, $F(9) = \frac{4}{1}$, and

$F(10) = \frac{5}{1}$. Then skip $\frac{4}{2}$, $\frac{3}{3}$, and $\frac{2}{4}$ (since $\frac{4}{2} = \frac{2}{1}$, $\frac{3}{3} = \frac{1}{1}$, and $\frac{2}{4} = \frac{1}{2}$), and set $F(11) = \frac{1}{5}$.

The Set of All Positive Rational Numbers Is Countable

Continue in this way, defining $F(n)$ for each positive integer n .

Note that every positive rational number appears somewhere in the grid, and the counting procedure is set up so that every point in the grid is reached eventually. Thus the function F is onto.

Also, skipping numbers that have already been counted ensures that no number is counted twice.

Thus F is one-to-one. Consequently, F is a function from $\mathbf{Z}^+$ to $\mathbf{Q}^+$ that is one-to-one and onto, and so $\mathbf{Q}^+$ is countably infinite and hence countable.

Uncountability of the set of all real numbers between 0 and 1

Theorem 7.4.2 (Cantor)

The set of all real numbers between 0 and 1 is uncountable.

The Search for Larger Infinities: The Cantor Diagonalization Process

Theorem 7.4.3

Any subset of any countable set is countable.

Corollary 7.4.4

Any set with an uncountable subset is uncountable.

The Search for Larger Infinities: The Cantor Diagonalization Process

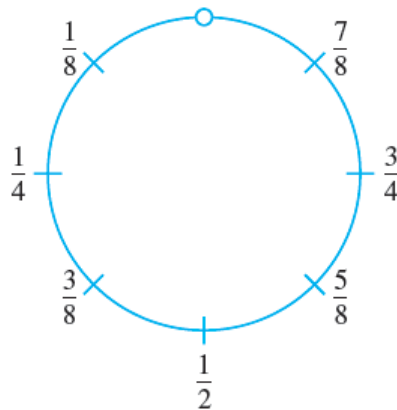
Show that the set of all real numbers has the same cardinality as the set of real numbers between 0 and 1.

The Search for Larger Infinities: The Cantor Diagonalization Process

Let S be the open interval of real numbers between 0 and 1:

$$S = \{x \in \mathbf{R} \mid 0 < x < 1\}.$$

Imagine picking up S and bending it into a circle as shown below.

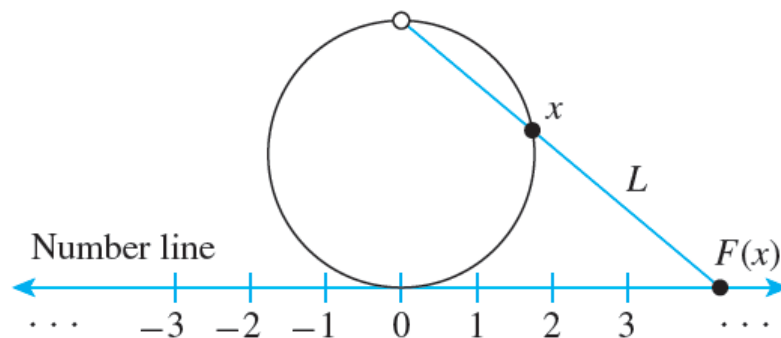


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Since S does not include either endpoint 0 or 1, the top-most point of the circle is omitted from the drawing.

Define a function $F: S \rightarrow \mathbf{R}$ as follows:

Draw a number line and place the interval, S , somewhat enlarged and bent into a circle, tangent to the number line at the point 0. This is shown below.



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For each point x on the circle representing S , draw a straight line L through the top-most point of the circle and x , and let $F(x)$ be the point of intersection of L and the number line. ($F(x)$ is called the *projection* of x onto the number line.)

It is clear from the geometry of the situation that distinct points on the circle go to distinct points on the number line, so F is one-to-one.

In addition, given any point y on the number line, a line can be drawn through y and the top-most point of the circle.

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This line must intersect the circle at some point x , and, by definition, $y = F(x)$.

Thus F is onto. Hence F is a one-to-one correspondence from S to $\mathbf{R}$, and so S and $\mathbf{R}$ have the same cardinality.

Application: Cardinality and Computability

Show that the set of all computer programs in a given computer language is countable.

Application: Cardinality and Computability

This result is a consequence of the fact that any computer program in any language can be regarded as a finite string of symbols in the (finite) alphabet of the language.

Given any computer language, let P be the set of all computer programs in the language. Either P is finite or P is infinite. If P is finite, then P is countable and we are done.

If P is infinite, set up a binary code to translate the symbols of the alphabet of the language into strings of 0's and 1's.

Application: Cardinality and Computability

(For instance, either the seven-bit American Standard Code for Information Interchange, known as ASCII, or the eight-bit Extended Binary-Coded Decimal Interchange Code, known as EBCDIC, might be used.)

For each program in P , use the code to translate all the symbols in the program into 0's and 1's.

Order these strings by length, putting shorter before longer, and order all strings of a given length by regarding each string as a binary number and writing the numbers in ascending order.

Application: Cardinality and Computability

Define a function $F: \mathbf{Z}^+ \rightarrow P$ by specifying that

$F(n)$ = the n th program in the list for each $n \in \mathbf{Z}^+$.

By construction, F is one-to-one and onto, and so P is countably infinite and hence countable.

As a simple example, suppose the following are all the programs in P that translate into bit strings of length less than or equal to 5:

10111, 11, 0010, 1011, 01, 00100, 1010, 00010.

Application: Cardinality and Computability

Ordering these by length gives

length 2: 11, 01

length 4: 0010, 1011, 1010

length 5: 10111, 00100, 00010

And ordering those of each given length by the size of the binary number they represent gives

$$01 = F(1)$$

$$11 = F(2)$$

$$0010 = F(3)$$

$$1010 = F(4)$$

Application: Cardinality and Computability

$$1011 = F(5)$$

$$00010 = F(6)$$

$$00100 = F(7)$$

$$10111 = F(8)$$

Note that when viewed purely as numbers and ignoring leading zeros, $0010 = 00010$.

This illustrates why it is important to order the strings by length before arranging them in ascending numeric order because otherwise the values of F would not be uniquely determined.

Application: Cardinality and Computability

- a. Let T be the set of all functions from the positive integers to the set $\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$. Show that T is uncountable.
- b. Derive the consequence that there are noncomputable functions. Specifically, show that for any computer language there must be a function F from $\mathbf{Z}^+$ to $\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$ with the property that no computer program can be written in the language to take arbitrary values as input and output the corresponding function values.

Application: Cardinality and Computability

- a. Let S be the set of all real numbers between 0 and 1. As noted before, any number in S can be represented in the form

$$0.a_1a_2a_3\dots a_n\dots,$$

where each a_i is an integer from 0 to 9. This representation is unique if decimals that end in all 9's are omitted.

Define a function F from S to a subset of T as follows:

$F(0.a_1a_2a_3\dots a_n\dots)$ = the function that sends each positive integer n to a_n .

Application: Cardinality and Computability

Choose the co-domain of F to be exactly that subset of T that makes F onto, recalling that T is the set of all functions from $\mathbf{Z}^+$ to $\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$.

In other words, define the co-domain of F to equal the image of F .

Now F is one-to-one because in order for the functions $F(x_1)$ and $F(x_2)$ to be equal, they must have the same value for each positive integer, and so each decimal digit of x_1 must equal the corresponding decimal digit of x_2 , which implies that $x_1 = x_2$.

Application: Cardinality and Computability

Thus F is a one-to-one correspondence from S to a subset of T . But S is uncountable by Theorem 7.4.2.

Theorem 7.4.2 (Cantor)

The set of all real numbers between 0 and 1 is uncountable.

Hence T has an uncountable subset, and so, by Corollary 7.4.4, T is uncountable.

Corollary 7.4.4

Any set with an uncountable subset is uncountable.

Application: Cardinality and Computability

b. Part (a) shows that the set T of all functions from $\mathbf{Z}^+$ to $\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$ is uncountable.

But, by Example 7.4.6, given any computer language, the set of all programs in that language is countable.

Consequently, in any computer language there are not enough programs to compute values of every function in T . There must exist functions that are not computable!